

Week 5 Tasks

OOP, Encapsulation, Exceptions, Debugging, File Handling, CSV, JSON

Task 1 — Student Class

Create class:

```
Student
```

Attributes:

- name
- roll
- marks

Methods:

- `grade()`
- `show_report()`

Task 2 — Bank Account Class

Create class:

```
BankAccount
```

Attributes:

- owner
- balance

Methods:

- `deposit(amount)`
- `withdraw(amount)`
- `show_balance()`

Add validation:

- amount must be positive
- withdrawal cannot be more than balance

Task 3 — Class Attribute Counter

Create class:

```
User
```

Add class attribute:

```
total_users
```

Every time a new user object is created, increase `total_users`.

Print total users at the end.

Task 4 — Encapsulated Employee Class

Create class:

```
Employee
```

Attributes:

- name
- private salary

Methods:

- `get_salary()`
- `set_salary(value)`

Salary cannot be negative.

Task 5 — `__str__()` Practice

Create class:

```
Book
```

Attributes:

- title
- author
- price

Add `__str__()` so printing the object gives neat output like:

```
Python Basics by Ali Khan - Rs. 1200
```

Task 6 — Inheritance Practice

Create:

```
Person
Teacher(Person)
```

Person has:

- name
- age

Teacher has:

- subject

Use `super()` properly.

Task 7 — Safe Division with Exceptions

Take two numbers from user.

Handle:

- invalid input
- division by zero

Use separate `except` blocks.

Task 8 — Marks Validator with `raise`

Create function:

```
set_marks(marks)
```

If marks are less than 0 or greater than 100, raise `ValueError`.

Call it using `try/except`.

Task 9 — Text File Student Report

Take student name, roll, and marks from user.

Save a report in:

```
student_report.txt
```

Example file content:

```
Student Report  
Name: Maria  
Roll: 101  
Marks: 92
```

Task 10 — JSON Student Records

Create a list of student dictionaries:

```
students = [  
    {"name": "Ali", "marks": 78},  
    {"name": "Maria", "marks": 92},  
    {"name": "Zain", "marks": 88}  
]
```

Save it to:

```
students.json
```

Then read it back and print each student neatly.